Applied AI & Machine Learning
CS-333

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PNEC, NUST

Lecture 5

Spring 2026

Correlations Between Variables

Learning Outcomes

By the end of this lecture, students will be able to:

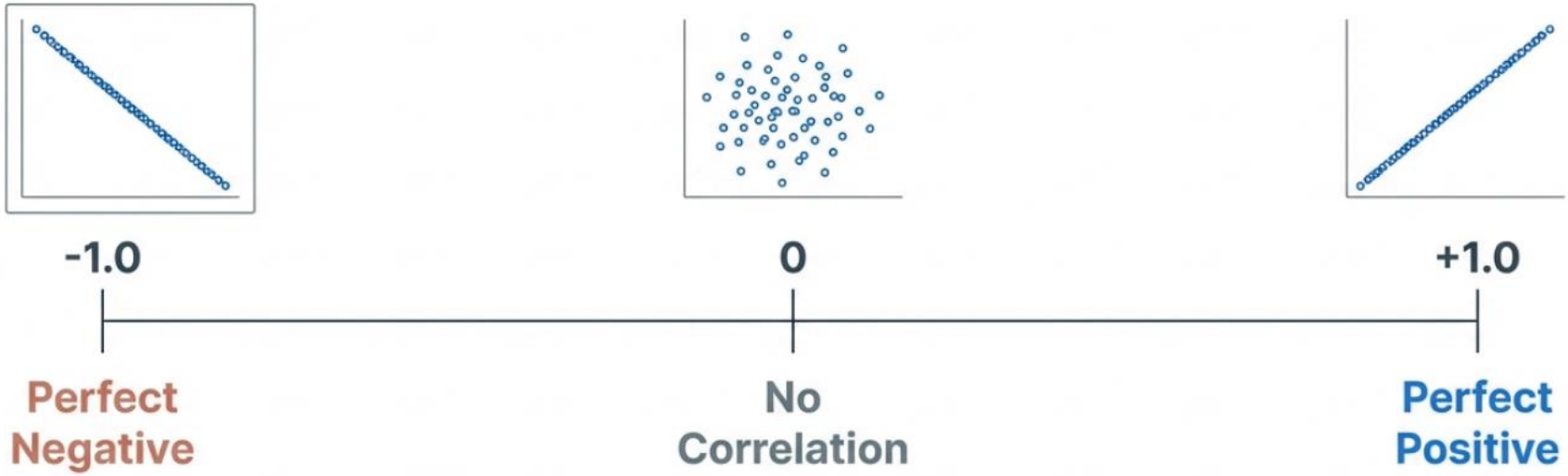
- Define correlation and its role in machine learning
- Calculate and interpret Pearson correlation coefficient
- Explain correlation strength and direction
- Distinguish correlation from causation
- Understand point-biserial correlation and its applications
- Apply correlation for feature selection and data analysis

Pearson Correlation Coefficient

Deciphering Linear Relationships

The Spectrum of Association

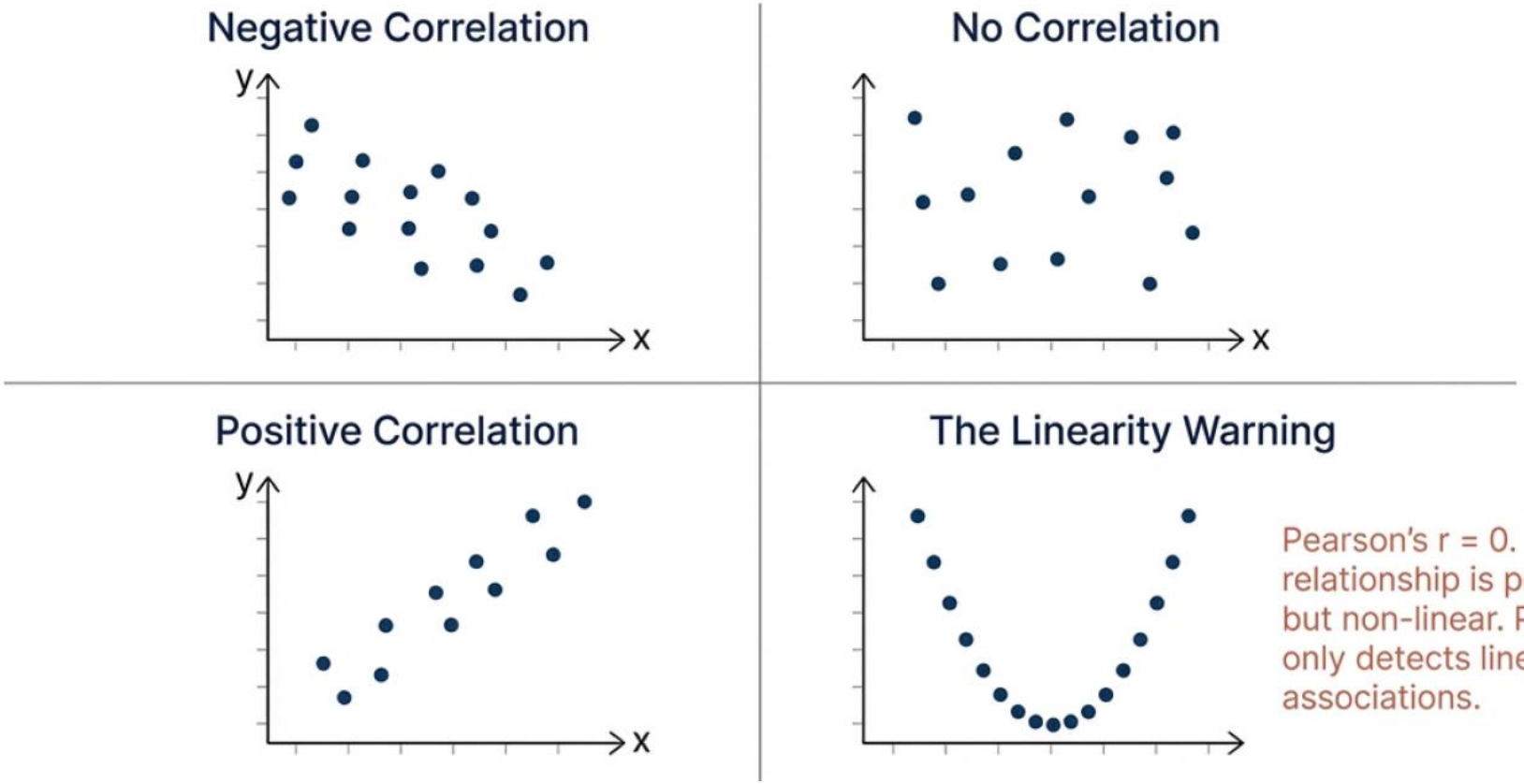
The Pearson Correlation Coefficient (*r*) quantifies the strength and direction of the linear relationship between two continuous variables.



Insight: Pearson's *r* is scale-independent. Unit changes (e.g., inches vs. centimeters) do not affect the coefficient.

Visualizing the Relationship

Calibrating the eye to data patterns.



Pearson's r = 0. This relationship is perfect but non-linear. Pearson only detects linear associations.

The Anatomy of the Formula

COVARIANCE

Measures if X and Y vary together. Do they move in the same direction relative to their means?

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

STANDARD DEVIATION

Normalizes the result. Ensures the final value stays between -1 and +1, regardless of units.

Parametric Measure: Essentially asking, 'Do these two variables deviate from their average in sync?"

The 5 Rules of Engagement

Prerequisites for a valid analysis.

1. Continuous Variables: Data must be interval or ratio level (e.g., height, test scores).

2. Linearity: The relationship must follow a straight line (verify with scatterplots).

3. No Outliers: Extreme values can disproportionately skew r.

4. Normality: Variables should be bivariately normally distributed.

5. Homoscedasticity: Variability in one variable remains consistent across values of the other.

Critical: Observations must be independent. One case cannot influence another.

Interpreting Strength Benchmarks based on Evans (1996).

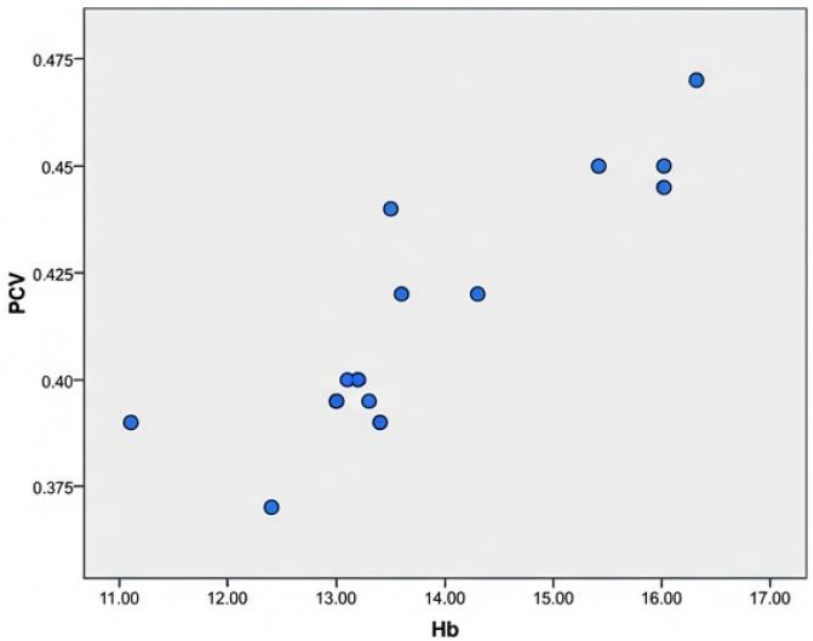
Range (r)	Strength
0.00 – 0.19	Very Weak
0.20 – 0.39	Weak
0.40 – 0.59	Moderate
0.60 – 0.79	Strong
0.80 – 1.00	Very Strong

Context Matters: These are general behavioral science guidelines. Physics or engineering may require much higher thresholds.

Reminder: Strength (magnitude) and Direction (sign) are distinct. An r of -0.90 is just as strong as +0.90.

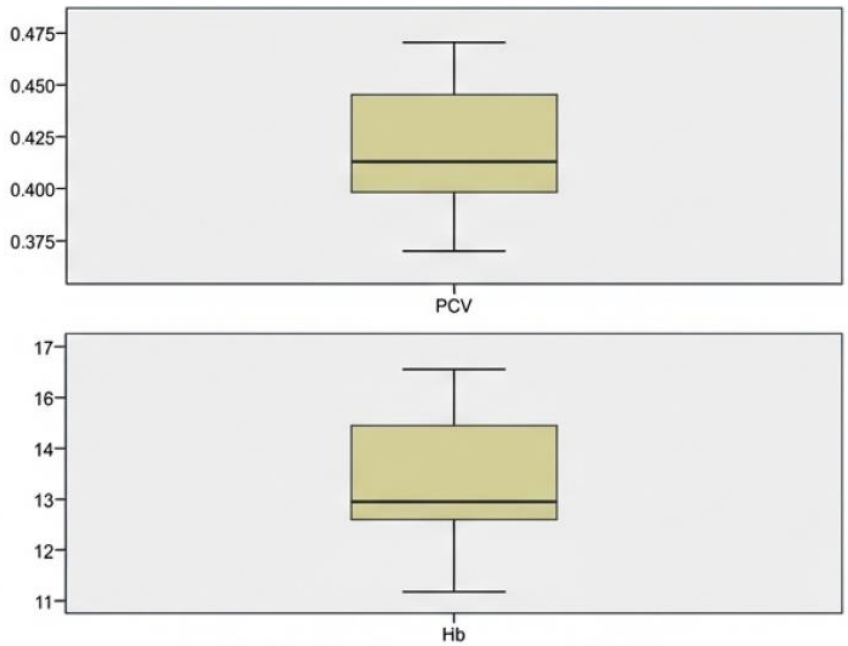
Case Study: Haemoglobin & PCV
Step 1: Visual Inspection & Assumption Checking

Visual Check: Linearity



Observation: Definite positive, linear relationship.

Assumption Check: Normality



Observation: Data is consistent with normal distribution; no major outliers.

Testing for Significance

Step 2: Statistical Calculation

The Hypotheses

Null (H_0): $\rho = 0$
(No correlation in population)

Alternative (H_1): $\rho \neq 0$
(Correlation exists)

The Results

$r = 0.877$
(Very Strong, Positive)

$p < .001$
(Significant)

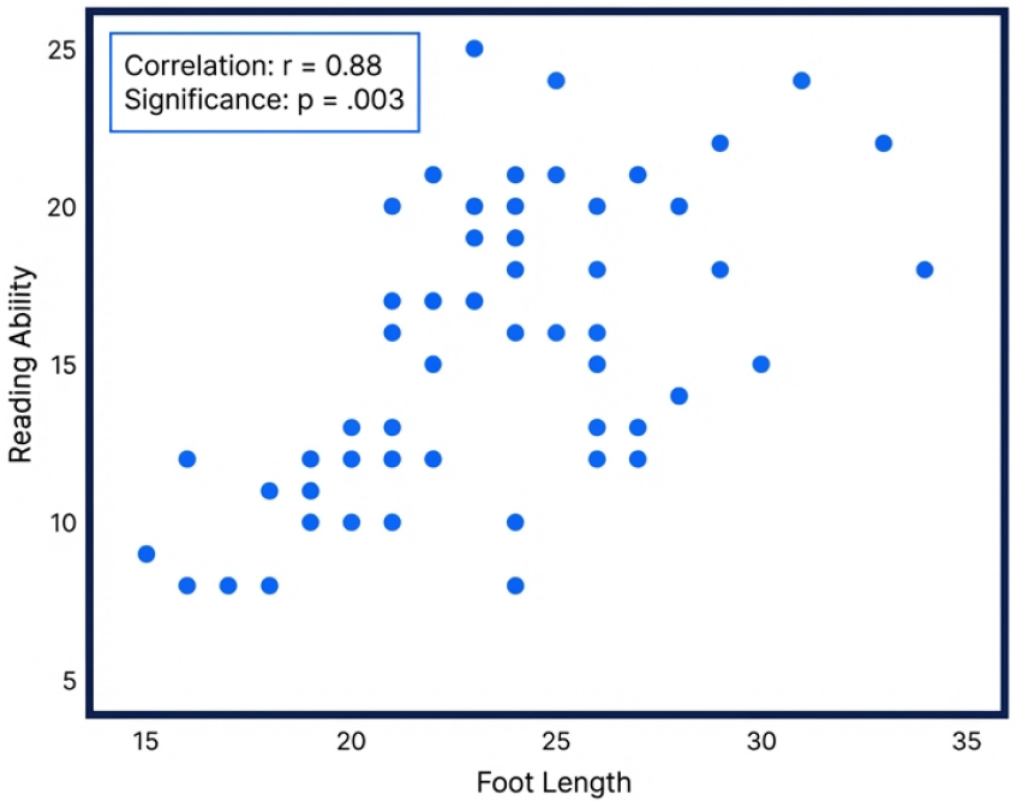
$N = 14$

The Report

There was a very strong, positive correlation between Hb and PCV ($r = .88$, $N=14$, $p < .001$). Large values of Hb are associated with large values of PCV.

The ‘Foot Length’ Trap

A Cautionary Tale



The Interpretation Trap:

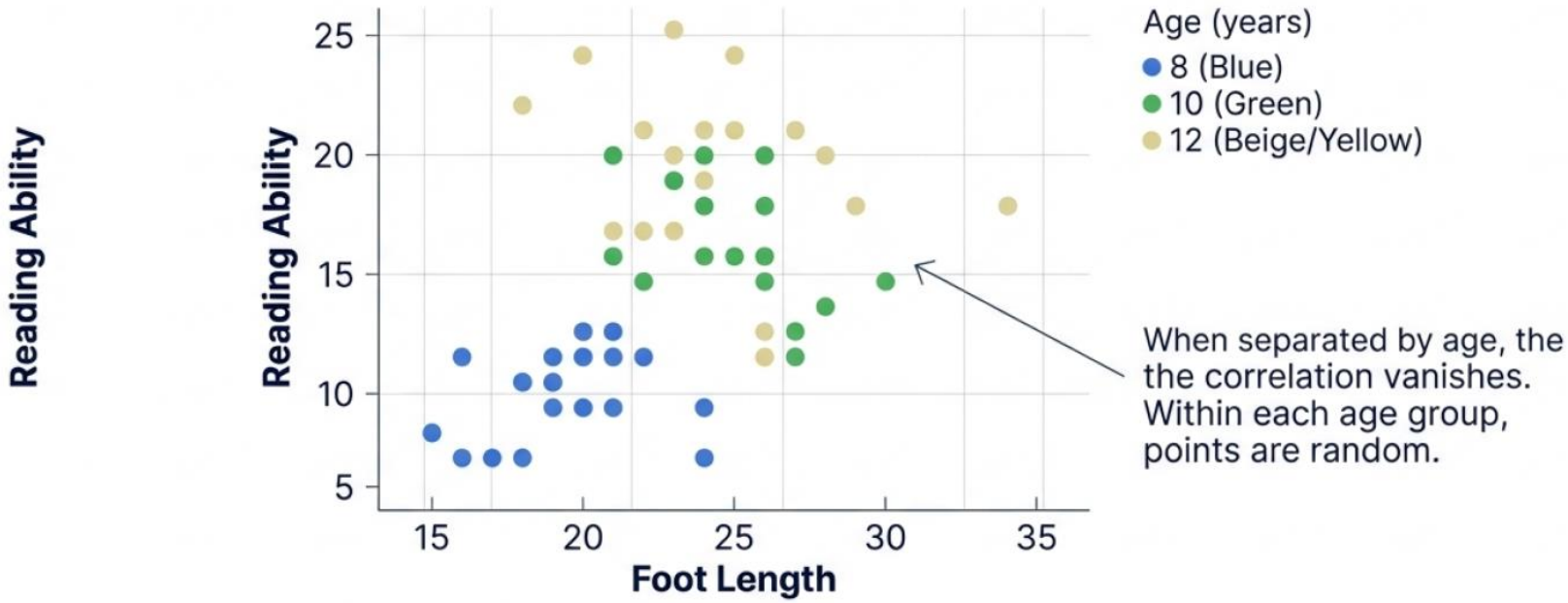
The statistics imply a very strong positive correlation.

Does having big feet make you a better reader?

Or does reading make your feet grow?

The Third Variable Problem

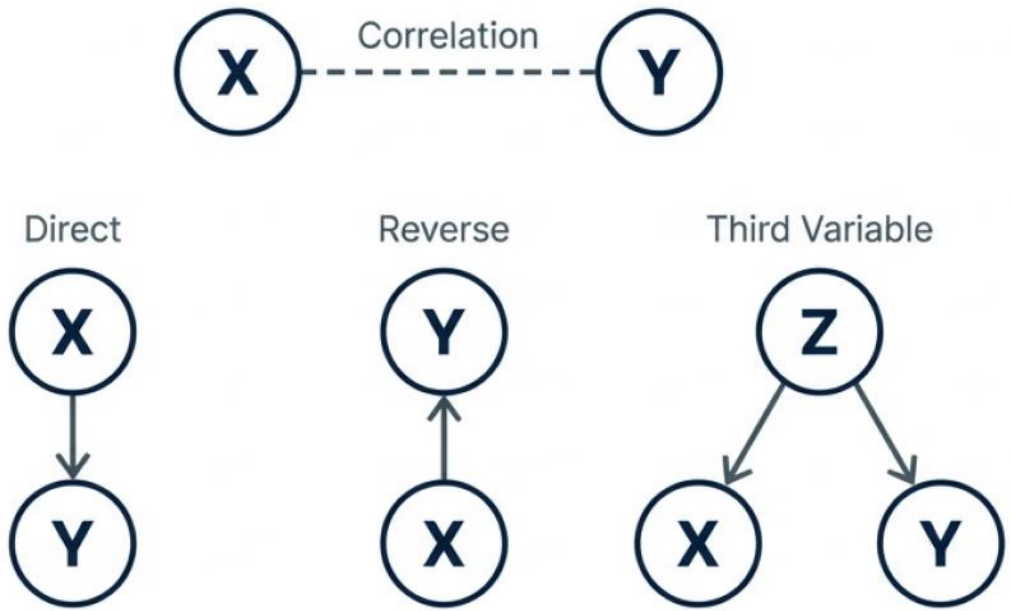
The Reveal: Stratification



Reality: **AGE** is the lurking variable. Older children have bigger feet feet AND better reading skills. This is a **spurious correlation**.

Correlation ≠ Causation

The Golden Rule of Statistics



Pearson *r* measures association, not mechanism. Even if *r* = 0.99, it does not prove that X causes Y.

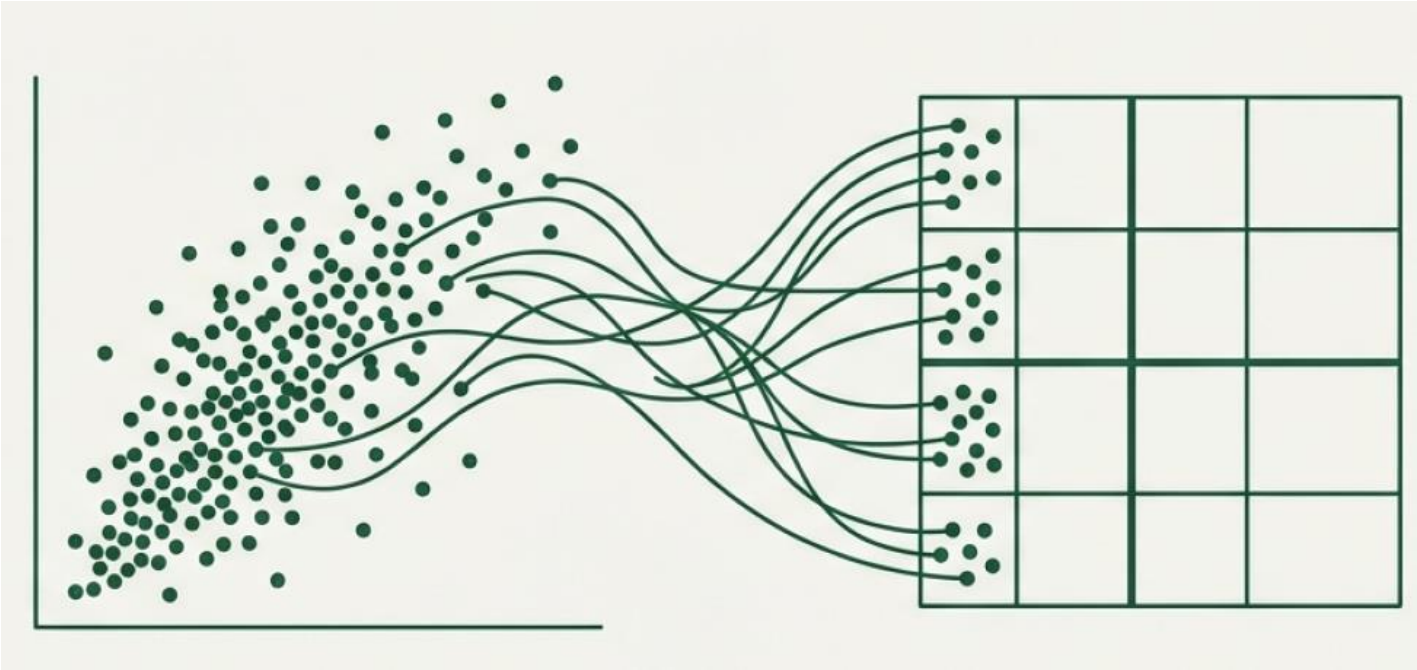
- Common Causes of Correlation:
  1. Direct Causation
  2. Reverse Causation
  3. Lurking Third Variables
  4. Coincidence

The Analyst's Checklist

1.	**Visualize First**	Always check the scatterplot for linearity and outliers before calculating.
2.	**Check Assumptions**	Ensure data is continuous, normally distributed, and independent.
3.	**Inspect the Matrix**	Ignore the main diagonal; focus on r (strength) and p (significance).
4.	**Contextualize**	Use standard guidelines (Weak/Moderate/Strong) but respect domain standards.
5.	**Be Skeptical**	Never assume causation. Always look for lurking variables (like Age).

Beyond Pearson: Measuring Association in Categorical Data

Point-Biserial Correlation



The Analyst's Decision Matrix

Continuous vs. Continuous

Pearson Correlation (r)

Categorical vs. Categorical

Chi-Square Test & Cramer's V

Binary vs. Continuous

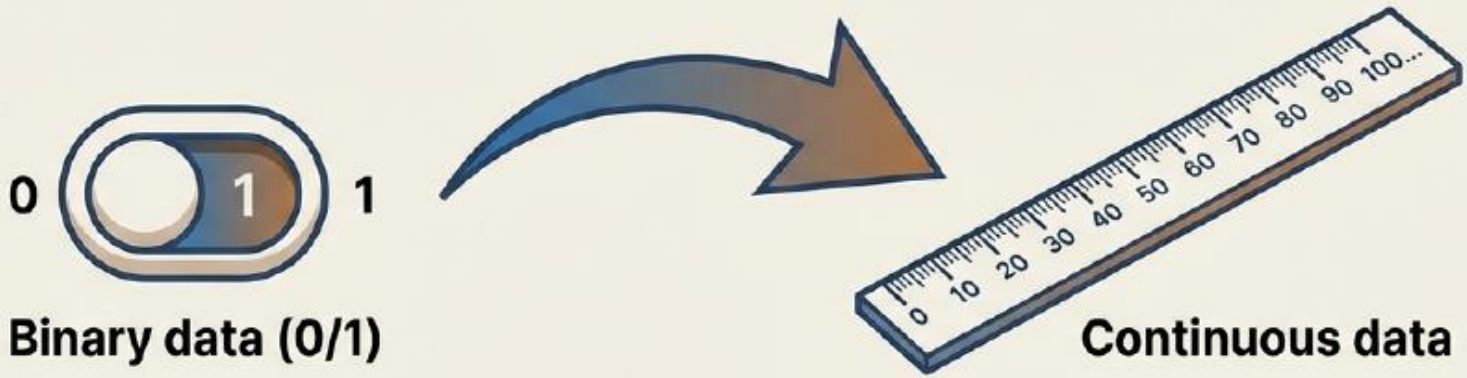
Point-Biserial Correlation (r_{pb})

Pearson is the standard for continuous metrics (e.g., Height vs. Weight).

However, real-world data is often categorical (Gender, Status, Pass/Fail).

We need a different toolkit for these frontiers.

Bridging the Gap: Point-Biserial Correlation



- **The Scenario:** Correlating one Binary Variable with one Continuous Variable.
- **Examples:** Gender (0/1) vs. Salary ($) OR Pass/Fail (0/1) vs. Hours Studied.
- **Definition:** A special case of Pearson correlation (r_{pb}).

Case Study: Do Study Hours Predict Success?

	Hours Studied (Metric)	Result (Binary)
1	10	Fail (0)
2	15	Pass (1)
3	12	Fail (0)
4	20	Pass (1)
...	...	...

- **Data Source:** (20 Students).
- **Transformation:** To run the math, categories are converted to numbers: Fail = 0, Pass = 1.
- **Question:** Does studying longer correlate with passing?

The Calculation (r_{pb})

Mean value of the persons who failed

Mean value of the persons who passed

$$r_{pb} = \frac{\bar{x}_2 - \bar{x}_1}{s_x} \cdot \sqrt{\frac{n_1 \cdot n_2}{n^2}}$$

Number of those who have falied

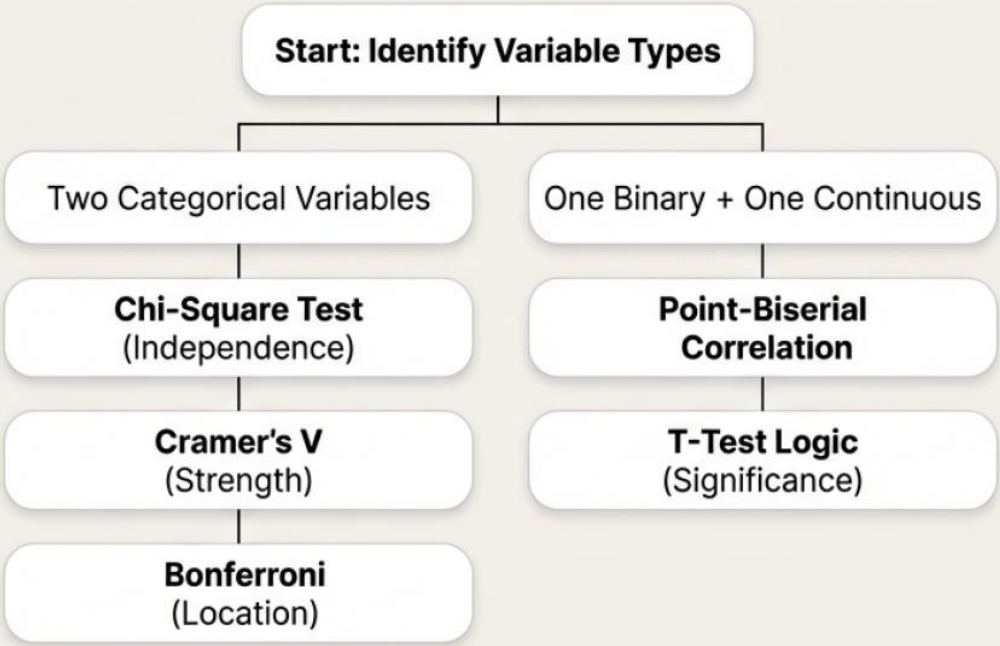
Number of people who have passed

Total number

$r_{pb} = 0.31$

1. **Positive Correlation**: More hours studied associates with passing.
2. **Magnitude**: 0.31 indicates a **Moderate** correlation.

The Analyst's Toolkit Summary



Data storytelling requires the right vocabulary. Choosing the correct test ensures your story is statistically sound.